

NACHA / ACH Payment System

Complete guide to US Automated Clearing House payment implementation

1. What is NACHA?

NACHA (National Automated Clearing House Association) is the organization that governs the ACH (Automated Clearing House) network in the United States. The ACH network is used to process billions of electronic financial transactions including payroll direct deposits, bill payments, consumer payments, and business-to-business payments.

ACH is a batch-processing system — transactions are collected throughout the day and processed in batches, typically overnight. Settlement usually occurs the next business day (T+1), though same-day ACH (SACH) is now available for an additional fee.

2. Key Participants

Participant	Acronym	Role
Originator	—	Entity that initiates the ACH payment (e.g., employer sending payroll)
Originating Depository Financial Institution	ODFI	The originator's bank; validates and submits batch files to ACH Operator
ACH Operator	—	NACHA-approved processors: Federal Reserve (FedACH) or The Clearing House
Receiving Depository Financial Institution	RFI	The receiver's bank; posts credit/debit to receiver's account
Receiver	—	Individual or business receiving the funds

3. SEC (Standard Entry Class) Codes

Each ACH transaction must specify an SEC code that describes the authorization type and entry method. The SEC code appears in the ACH file header and determines what information must be included in the transaction.

Code	Full Name	Use Case	Authorization
PPD	Prearranged Payment & Deposit	Consumer payroll, rent, utilities	Written authorization (paper)
CCD	Corporate Credit or Debit	B2B payments, vendor pay	Authorization agreement
WEB	Internet-Initiated Entry	Online bill pay, subscription	Online click-wrap agreement

Code	Full Name	Use Case	Authorization
TEL	Telephone-Initiated Entry	Phone payments	Recorded verbal authorization
CTX	Corporate Trade Exchange	EDI-formatted B2B payments	Trading partner agreement
IAT	International ACH Transaction	Cross-border payments	OFAC screening required

4. NACHA File Format

A NACHA file is a fixed-width text file (94 characters per line). It contains a File Header, one or more Batches (each with Batch Header, Entry Detail records, and Batch Control), and a File Control record.

■ File Structure

Line 1: 1 – File Header Record (94 chars)

Line 2: 5 – Batch Header Record (per batch)

Line 3-N: 6 – Entry Detail Records

Line N+1: 8 – Batch Control Record

Last: 9 – File Control Record

Example Entry Detail (Type 6):

6221234567890987654321 000015000JOHN DOE 0000001234567890

■ Key Fields in Entry Detail (Type 6)

Position	Length	Field	Description
1	1	Record Type	Always "6"
2	2	Transaction Code	22=Checking Credit, 27=Checking Debit, 32=Savings Credit
4	9	RDFI Routing	9-digit ABA routing number of receiver's bank
13	17	Account Number	Receiver's account number (right-padded with spaces)
30	10	Amount	Dollars + cents, no decimal (e.g., \$150.00 = 0000015000)
40	22	Individual Name	Receiver's name
63	15	Trace Number	8-digit ODFI routing + 7-digit sequence

5. K Star NACHA Implementation

K Star implements a simplified ACH payment service that models the NACHA payment lifecycle. Users can initiate ACH payments by providing originator and receiver bank details. The service generates NACHA-compliant identifiers (payment ID, trace number, batch number) and simulates the T+1 settlement process asynchronously using Spring @Async.

■ Payment Status Lifecycle

Status	Meaning	When
PENDING	Payment submitted, not yet processed	Immediately after API call

Status	Meaning	When
SUBMITTED	Batch submitted to ACH operator	~2 seconds (simulated ODFI processing)
PROCESSING	ACH operator is routing payment	During batch window
SETTLED	RDFI posted credit/debit to receiver	T+1 next business day
RETURNED	RDFI returned with return code	If account closed, NSF, unauthorized
FAILED	Payment could not be processed	Validation failure, invalid routing number

■ Common NACHA Return Codes

Code	Reason	Action Required
R01	Insufficient Funds	Notify originator; retry after funds available
R02	Bank Account Closed	Update receiver account information
R03	No Account/Unable to Locate	Verify routing and account numbers
R04	Invalid Account Number	Correct account number format
R06	Returned per ODFI Request	ODFI instructed return
R09	Uncollected Funds	Retry after hold period
R10	Customer Advises Not Authorized	Stop future transactions; investigate
R29	Corporate Customer Advises Not Authorized	Stop CTX entries; investigate